The 50 cm rule for SPD wiring

Why the wires to a surge protection device must be short, and how to keep them short

In one sentence

During a surge the connecting wires add their own voltage, about **1,000 V for every metre**, and that extra voltage lands on the equipment the SPD is meant to protect.

1 Why wire length matters

A surge is fast

At normal 50 Hz a short wire is almost a perfect conductor. A surge rises in a few millionths of a second, and at that speed every wire resists the change in current. This is inductance: about 1 µH per metre of panel wiring.

The voltage is large

The standard test surge (8 kA, rising in 8 µs) changes at about 1,000 A per microsecond.

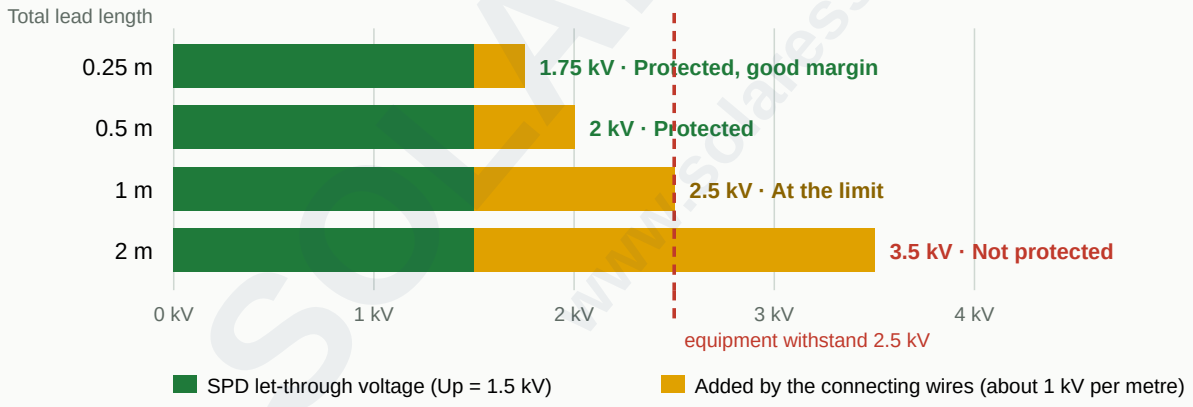
1 µH × 1,000 A/µs = 1,000 V per metre

It adds to the SPD's own

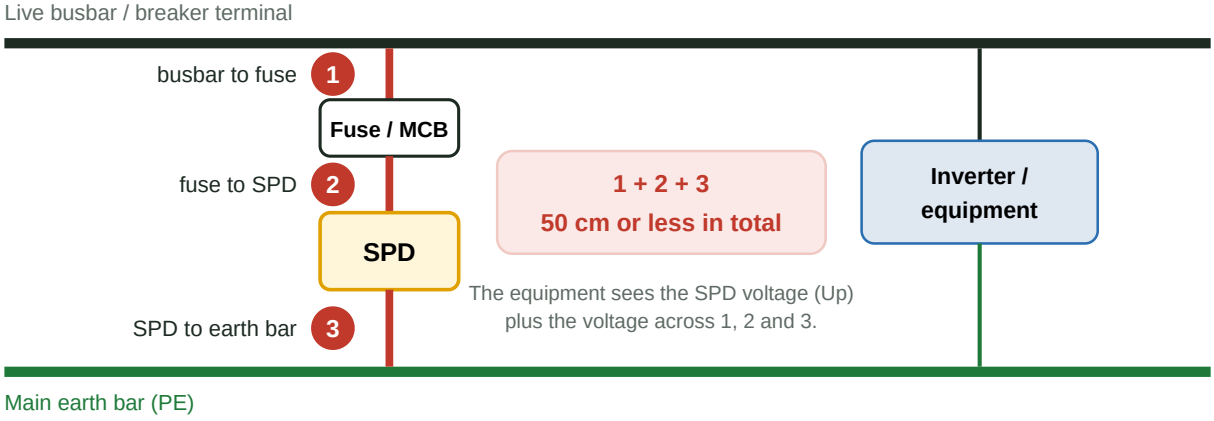
The equipment sees the SPD's let-through voltage (Up) **plus** the voltage across the connecting wires. Long wires can add more than the SPD itself lets through.

2 What it does to the protection

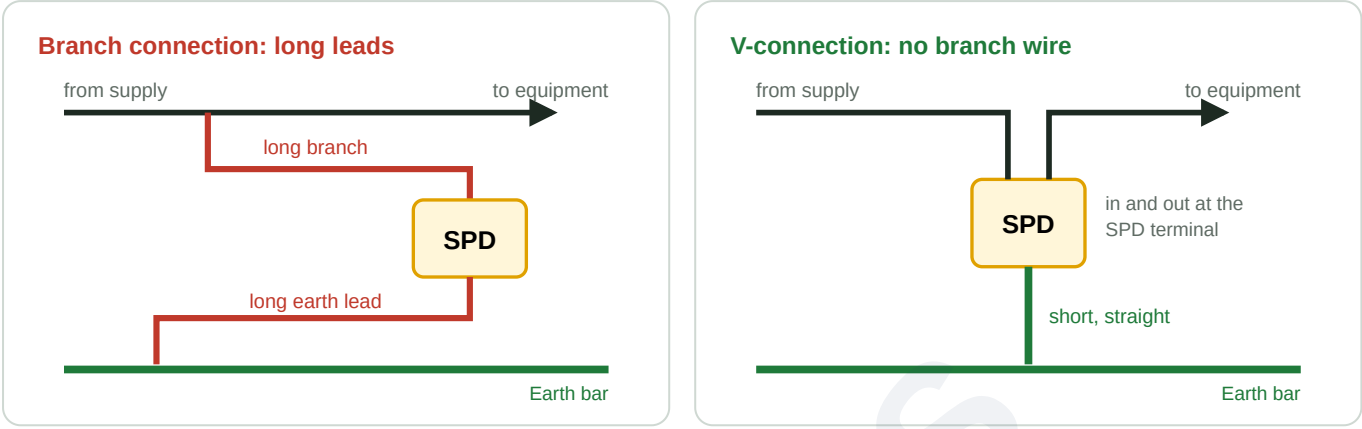
Example: a Type 2 SPD with Up = 1.5 kV protecting an inverter that withstands 2.5 kV. With 2 m of leads the wiring adds more voltage than the SPD itself lets through.



3 What counts toward the 50 cm



4 How to keep it short in a real board



- ☑ **Mount the SPD next to the earth bar:** not wherever there is free DIN rail.
- ☑ **Use a V-connection:** bring the live wire in and out at the SPD terminal, so there is no branch wire on the live side.
- ☑ **Add a local earth terminal:** if the main earth bar is far away, fit one beside the SPD and bond it to the main bar and to the equipment earth.
- ☑ **Use the metal enclosure as the earth path:** where the board maker allows it. A wide steel plate has far lower inductance than a wire.
- ☑ **Keep the wires straight and together:** no loops, no coiled spare length; run the live and earth leads side by side.

Thicker cable does not fix it

Cable size changes resistance, which is not the problem. Inductance barely changes with thickness. Use the minimum size (4 mm² copper for Type 2, 16 mm² for Type 1), then keep it short.

5 Site check

Measure three lengths

- Busbar or breaker terminal to the backup fuse
- Fuse to the SPD
- SPD to the main earth bar

Target

50 cm or less

in total. Treat 1 m as the absolute maximum.

If it is longer

Move the SPD, use a V-connection, or add a local earth terminal. Do not accept it as built.

Sources

Schneider Electric Electrical Installation Guide, "Connection of Surge Protection Device" (50 cm total length; about 1,000 V per metre for an 8 kA 8/20 µs surge; 4 mm² and 16 mm² minimum conductor sizes) · IEC 60364-5-534 (0.5 m preferred, 1 m maximum). The 1.5 kV and 2.5 kV figures in section 2 are a typical example, not values for a specific product: use the Up of the SPD and the withstand voltage of the equipment actually installed.

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